Do Sparse Autoencoders Capture Linguistic Typology?

Anonymous ACL submission

Abstract

Sparse Autoencoders (SAEs) have recently been used to uncover interpretable, language-specific features in multilingual large language models (LLMs). The SAE-LAPE framework (Andrylie et al., 2025) identifies such features using activation entropy and demonstrates their causal relevance for multilingual behavior. However, the linguistic nature of these features remains less clear.

In this study, we extend the SAE-LAPE analysis by introducing a typological probing stage using *lang2vec*. We investigate whether the language-specific SAE features discovered within Llama 3.2 correspond to known phonological, morphological, and syntactic relationships among languages. By correlating feature-sharing similarity across languages with typological distances from *lang2vec*, we aim to ground sparse features in established linguistic knowledge.

Our experiments reveal that neurons highly correlated (high R^2) with typological data are mostly tied to surface properties: phonological inventories, genealogical family membership, and basic word-order syntax. In contrast, neurons identified as specific to one language (or a few languages) largely persist even when input text is randomized, implying they encode superficial cues (e.g. common tokens or orthography) rather than grammar. Neuron sets shared across many languages show lower overlap under shuffling, suggesting they may capture more abstract structure. These findings illustrate a gap between neuron-level interpretability and true cross-linguistic generalization. Can we propose using typological alignment as a diagnostic benchmark to guide future probing and steering methods toward genuine linguistic features?

1 Introduction

Multilingually pretrained large language models (MLLMs) achieve strong multilingual performance, but their internal mechanisms remain difficult to interpret. Previous neuron-level studies have shown that multilinguality often relies on distributed representations, where multiple languages share overlapping sets of activations or "neurons". This superposition limits interpretability, making it hard to identify how specific linguistic concepts are represented within the model.

Recent work by Andrylie et al. (2025) introduced **Sparse Autoencoder Language Activation Probability Entropy (SAE-LAPE)**, a method that identifies *language-specific* features in multilingual LLMs. These features were found to be interpretable, localized in middle-to-deep layers, and causally linked to language behavior. However, the analysis remained largely model-internal: it did not examine whether these features align with established linguistic knowledge.

This study builds upon the SAE-LAPE framework by introducing an external **typological probing** component using the *lang2vec* database. Our goal is to determine whether the discovered sparse features correspond to known linguistic properties such as word order, morphology, or phonological systems. By comparing feature-sharing patterns with typological distances, we provide an interpretable link between internal model features and external linguistic typology.

Our approach is exploratory in nature. We treat each SAE feature (or neuron) as a variable whose activations across languages can be correlated with typological patterns. If certain features systematically predict aspects of phonology, syntax, or morphology, it would suggest that multilingual LLMs implicitly learn to organize languages along typologically consistent axes. We further validate these findings through ablation experiments on high-correlation features.

Research Questions This study addresses the following research questions:

- **RQ1:** Do the language-specific or multilingual sparse features discovered by SAE-LAPE align with typological dimensions defined in *lang2vec*? In other words, can a feature's activation profile across languages predict typological properties such as word order or phoneme inventory size?

- **RQ2:** Are certain layers or types of SAE features (e.g., language-specific vs. shared) more predictive of linguistic typology? Does typological alignment vary across model depth?

- **RQ3:** What is the causal role of typology-aligned features? Does ablating these features lead to selective degradation in performance for languages that share the corresponding typological trait?

Together, these questions seek to uncover whether sparse, interpretable components within multilingual LLMs reflect structured linguistic knowledge or arise purely as artifacts of data-driven optimization.

2 Methodology

2.1 Overview

We extend the SAE-LAPE analysis of Andrylie et al. (2025) by introducing a typological probing framework. Our goal is to determine whether language-specific or multilingual sparse features identified by SAE-LAPE align with linguistic typological dimensions obtained from *lang2vec* ().

We adopt a regression-based probing approach, where each neuron (or sparse feature) is treated as a variable with activation statistics across languages, and the target variables are the corresponding typological attributes of those languages.

2.2 Data and Feature Extraction

Model and Sparse Autoencoders. We use the SAE features extracted from the Llama 3.2 1B model via the publicly available TopK-SAE checkpoints from EleutherAI.[1] Each feature represents a learned sparse direction within the feed-forward networks (FFNs).

Following Andrylie et al. (2025), we first compute the average activation of each SAE feature for every language in a multilingual evaluation set consisting of 15 typologically diverse languages (English, French, Spanish, Russian, Japanese, Korean, Arabic, Hindi, etc.).

This produces an activation matrix $A \in \mathbb{R}^{F \times L}$, where F is the number of SAE features and L the number of languages. Each row corresponds to one feature's activation profile across languages.

Language Classification of Features. As in SAE-LAPE, we categorize each feature according to its activation entropy across languages:

- **Language-specific:** low-entropy features strongly active for one or two languages.

- **Multilingual:** features active across several languages with moderate entropy.

- **Universal:** high-entropy features active for most languages.

This classification is used later to compare typological predictiveness across feature types.

2.3 Typological Data and Feature Selection

We use the *lang2vec* library () to extract typological representations for each language. Lang2vec aggregates linguistic features from multiple authoritative sources, including the World Atlas of Language Structures (WALS), PHOIBLE, SSWL, Ethnologue, and Glottolog, and provides consistent vector representations across over 2000 languages.

Each language is represented as a vector $T_l \in \mathbb{R}^D$, where D corresponds to the concatenation of all available typological dimensions.

Syntax-related features:

- **syntax_wals**: Word order and syntactic structures from WALS (e.g., SVO/SOV order, noun-adjective position, subject–verb agreement).

- **syntax_sswl**: Structural features from the SSWL database, covering morphological and clausal variation.

- **syntax_ethnologue**: Typological summaries drawn from Ethnologue.

- **syntax_knn**: Nearest-neighbor syntactic distance features computed via KNN-based typological inference.

- **syntax_average**: Mean-pooled syntactic embeddings combining multiple databases.

[1] https://huggingface.co/EleutherAI/sae-Llama-3.2-1B-131k

Phonology-related features:

- **phonology_wals**: Sound system and phonological process features (e.g., tone, stress, syllable structure).
- **phonology_ethnologue**: Phonological summaries derived from Ethnologue.
- **phonology_knn**: Phonological nearest-neighbor features.
- **phonology_average**: Averaged phonological features across sources.

Phoneme inventory features:

- **inventory_ethnologue**: Phoneme inventory statistics from Ethnologue.
- **inventory_phoible_aa, gm, saphon, spa, ph, ra, upsid**: Seven datasets from PHOIBLE and related sources, each providing complementary phoneme inventory information across different regional databases.
- **inventory_knn**: KNN-based phoneme inventory similarity features.
- **inventory_average**: Mean-pooled inventory feature representation.

Additional metadata:

- **fam**: Binary indicators for genealogical language family membership (e.g., Indo-European, Afro-Asiatic, Sino-Tibetan).
- **geo**: Normalized latitude and longitude coordinates for geographic proximity.
- **id**: Language identifiers used for cross-matching with model activations.

This results in a typological feature matrix $T \in \mathbb{R}^{L\times D}$, where each row corresponds to a language and each column to one of the above typological attributes. All continuous features are z-normalized to zero mean and unit variance, and missing values are imputed with the column mean.

By maintaining all feature classes, we capture both fine-grained linguistic structures (e.g., PHOIBLE inventory traits) and coarse typological similarities (e.g., syntactic averages and family membership), allowing us to test which domains align most strongly with the internal SAE representations.

2.4 Regression-Based Typological Probing

For each SAE feature f, we consider its activation vector across languages $a_f \in \mathbb{R}^L$ and fit a linear regression model to predict the typological features:

$$T = a_f W + \epsilon$$

where W is the regression coefficient vector and ϵ is residual noise.

We evaluate the regression using the coefficient of determination (R^2 score). This R^2 score reflects how much of the typological variance across languages can be explained by the activation pattern of a single SAE feature.

We repeat this for all features and analyze:

- The distribution of R^2 scores across features.
- Differences between language-specific, multilingual, and universal features.
- Which typological domains (syntax, phonology, morphology) yield the highest predictability.

2.5 Layer-Wise Analysis

To investigate where typological structure emerges in the model, we compute mean R^2 per layer. This identifies whether shallow, middle, or deep layers capture linguistic typology most strongly. We hypothesize that phonological correlations appear in shallow layers, while syntactic and morphological ones dominate middle-to-deep layers, consistent with hierarchical processing patterns reported by Andrylie et al. (2025).

2.6 Shuffling Ablation

We create a "shuffled" version of the corpus for each language by randomly permuting words, thus destroying syntactic order but preserving vocabulary distribution. We rerun the SAE-LAPE identification on these shuffled inputs to get a new set of language-specific neurons. We then measure the overlap between the original and shuffled neuron sets for each category of sharing. Concretely, for neurons labeled specific to k languages (or just one), we compute the Jaccard similarity of the original vs. shuffled sets. A high Jaccard means the same neurons were picked even when word order was lost, implying those neurons did not rely on grammar. We expect that neurons truly encoding syntax should have low overlap under shuffling, whereas neurons capturing surface cues (e.g. script or frequent words) would overlap strongly.

2.7 Romanization Ablation

For languages written in non-Latin scripts South Asian, we transliterate text into the Latin alphabet. For this, we have used Google's Dakshina dataset. We repeat the SAE-LAPE analysis on the romanized text and compare which neurons are identified. This tests whether some neurons are tied to orthographic form: if romanization causes a large change in the neuron set, then many language-specific neurons may have been sensitive to script rather than phonology or syntax.

2.8 Activation steering

Finally, we perform exploratory ablation analysis on the top typology-correlated features. For selected languages, we zero out activations of high-R^2 features and measure changes in perplexity or output coherence. This helps validate whether typologically aligned features have a functional role in maintaining language-specific behavior.

2.9 Evaluation Summary

Our evaluation combines three complementary analyses:

1. **Predictive probing:** R^2 regression analysis between SAE activations and typological vectors.

2. **Layer-wise trends:** Aggregating typological predictiveness per model layer.

3. **Causal validation:** Feature ablation to assess the behavioral impact of typology-aligned features.

Together, these provide a quantitative and interpretable view of how linguistic typology is represented within sparse features of multilingual LLMs.

3 Results

3.1 Typological Correlations

We find that the most predictive features for neuron activations come from phonological inventories, genealogical family, and basic syntax. For example, neurons show strong alignment (high average and maximum R^2 scores) with one-hot family indicators (e.g. Indo-European vs. others). Certain WALS syntax features (e.g. basic word order) also appear with moderate R^2, but often less so than inventory features. In contrast, purely phonetic features tend to produce weak correlations. These trends are summarized in Table 1. Notably, the highest-R^2 alignments overwhelmingly belong to the "syntax" category (e.g. word order) and the "fam" category (language family membership).

Interestingly, we observe different kinds of trends when we observe the maximum R^2 per neuron vs. its average over all dimensions of the feature set.

3.1.1 Layerwise trends

- When we look at the **average** R^2 scores (see fig 1) across all dimensions of every feature set in the family, there is a consistent trend of `syntax` > `fam` > `inventory` > `phonology`.

- In this case the, the first three features maximally aligned features become more aligned in the middle layers, whereas the pattern is opposite for the least aligned feature.

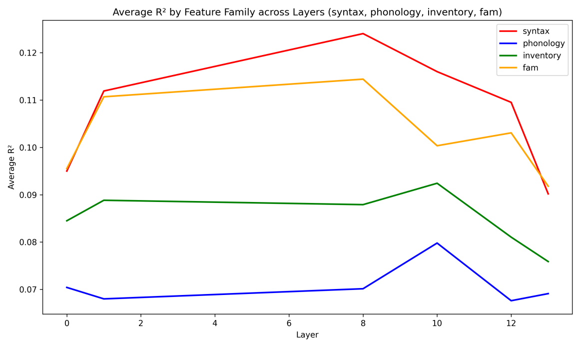


Figure 1: Average R^2 by feature family, against layer index

- Another line of analysis is looking at the maximum R^2 score for every feature set, and then averaging them. A very high score of any neuron for any dimension of one feature set indicates a good predictive correlation for at least some sub-feature captured by the feature set. In this case, the maximum R^2 patterns are somewhat similar in order (see 2). Here `fam` is the highest (almost 1.0), `syntax` feature is a close second.

- Here, the maxima dip in the middle layers (as opposed to the average R^2 scores above).

3.1.2 Based on level of specificity

- We have plotted the R^2 patterns across degree of sharedness (number of shared languages).

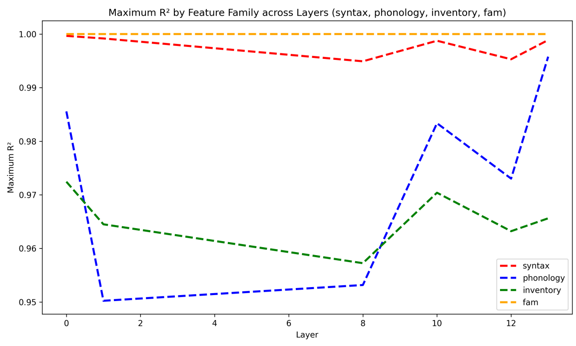


Figure 2: Maximum R^2 by feature family, against layer index

Again, both cases (average and maximum are analysed).

- For the average case (see 3) we see no clear patterns accept that the overall order is loosely maintained.
- For the maximum case (see 4), there is a very clear pattern of the maximum score cropping as the degree of sharedness increases, with the overall order holding. This indicates that the SAE-LAPE method tends to select neurons which are "well-aligned typologically with some sub-features encoded by lang2vec".

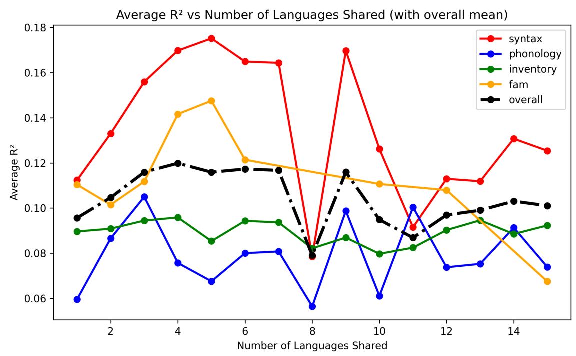


Figure 3: Average R^2 scores against degree of sharedness

3.1.3 Per-layer R^2 distributions

3.2 Shuffling Ablations

3.2.1 Neuron Set Overlaps

- We observe that single-language neuron sets have very high overlap, meaning that most neurons flagged as specific to one language remain the same after shuffling word order.
- As the number of languages a neuron is shared across increases, the Jaccard similarity systematically decreases first, then increases again (as the number of languages increase, the shared neuron set tends to overlap).

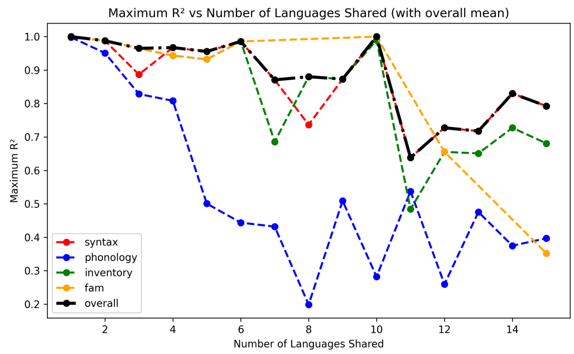


Figure 4: Maximum R^2 scores against degree of sharedness

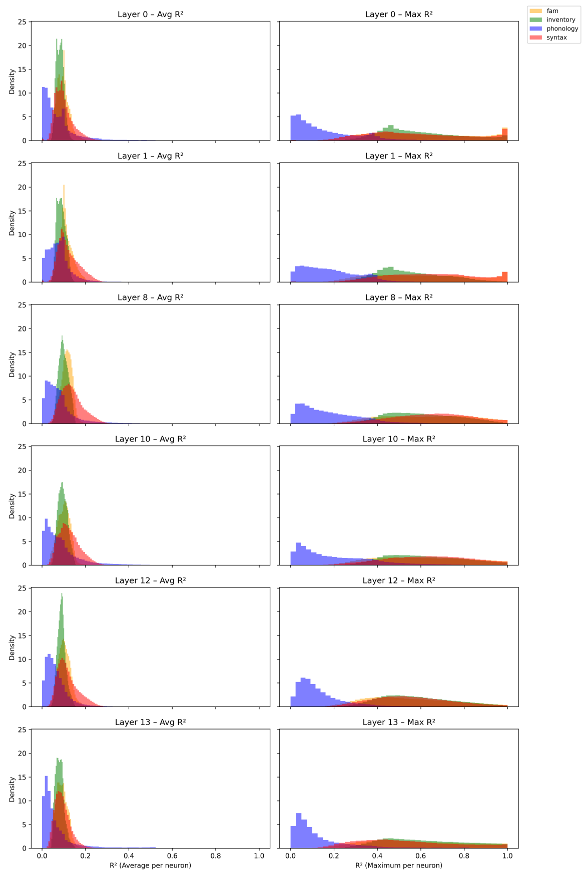


Figure 5: R^2 distributions for all SAE neurons across layers. Note how we have consistent distribution patterns which vary among the feature sets.

3.2.2 Distribution Analysis

Contrary to what should be ideally expected, there is no statistically significant shift in the syntax-R^2 distribution of the non-overlapping neurons as compared to the overlapping neurons, indicat-

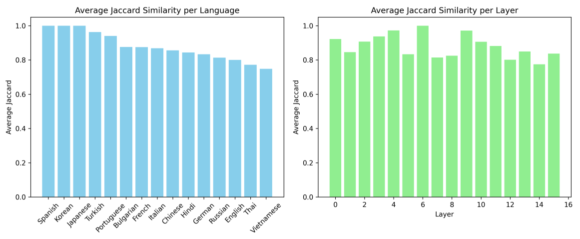


Figure 6: Jaccard similarities between SAE neuron sets identified as specific to 1 language, before and after the example sentences are shuffled.

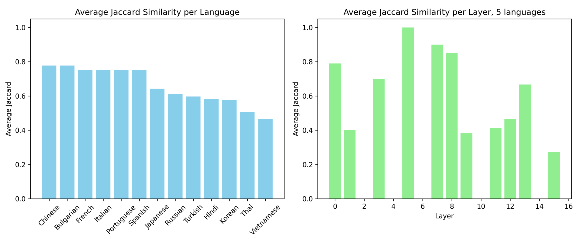


Figure 7: Jaccard similarities between SAE neuron sets identified as specific to 5 languages language, before and after the example sentences are shuffled.

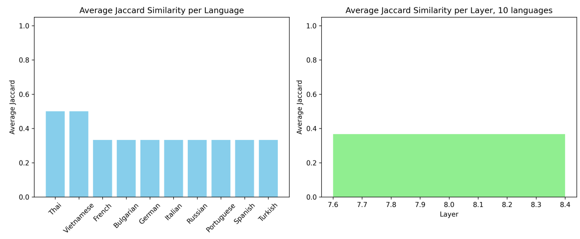


Figure 8: Jaccard similarities between SAE neuron sets identified as specific to 10 languages language, before and after the example sentences are shuffled.

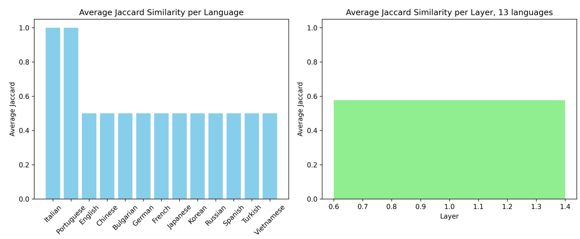
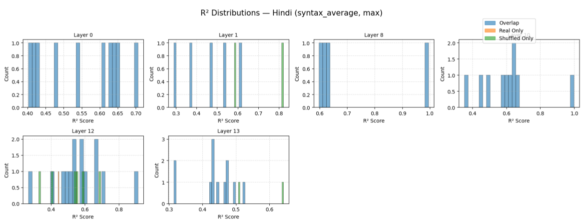


Figure 9: Jaccard similarities between SAE neuron sets identified as specific to 13 languages language, before and after the example sentences are shuffled.

ing that, on average, the neurons don't align with the `syntax` typology. If the non-overlapping neurons don't correspond to syntax, it means that the language-specific neurons don't capture syntactic cues. These are, probably captured by other neurons at different levels of sharing.



3.3 Romanization Ablation

Contrary to what one might hope if these neurons capture true language identities, romanization dramatically changes the picture. For every non-Latin script language tested, the Jaccard similarity between the feature sets from native vs. romanized text is effectively zero. In other words, almost none of the features identified as "Arabic-specific" on the Arabic script data match those identified on the romanized Arabic. This is despite using the same tokenizer for both experiments. Qualitatively, the features from native text often correspond to specific Unicode character clusters or script-specific phonetic cues, whereas the romanized features reflect the Latin spelling patterns.

4 Deeper investigation of typological alignment

A deeper dive into the distributions of the alignment scores (R^2) shows that distinctions between all pairs of languages are not learnt at similar levels.

4.1 Strong hierarchical recoverability of typology

R² scores follow a stable ordering across layers: `Family` $>$ `Inventory` $>$ `Syntax` $>$ `Phonology` $>$ `Geography`

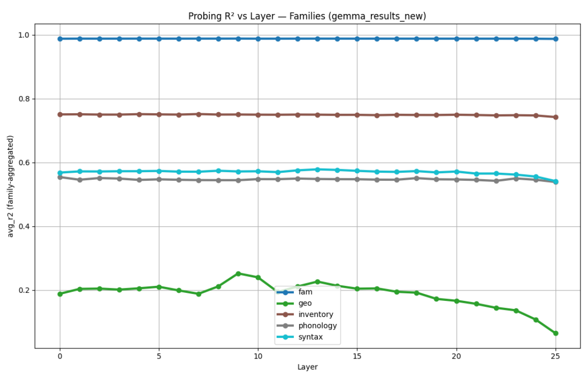


Figure 10: Ordering of R^2 scores for Gemma-2-2B. There is a strong significant ordering in the average scores of different typological properties.

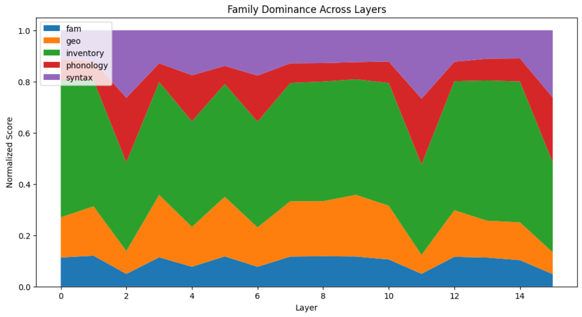


Figure 11: Family dominance across layers for Llama-1B-SAEs

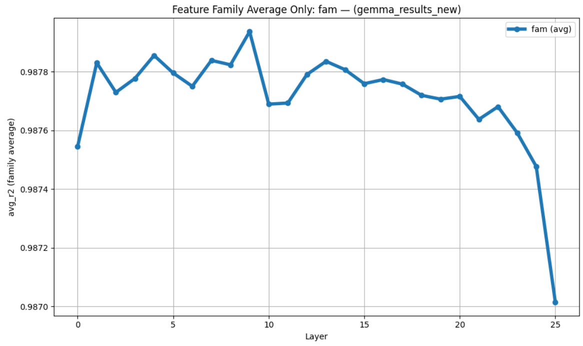


Figure 12: Gemma SAEs: trend of scores for `fam` feature

We can draw the following interpretation from these patterns.

- Language family (coarse, categorical) is the dominant signal in the multilingual activation space.
- Inventory features (phoneme/orthography) are highly recoverable due to strong token-level statistical patterns.
- Syntax is moderately encoded but diluted in averaged activations.
- Phonology is weakly encoded because it is only indirectly visible.
- Geography is minimally encoded, as expected for a non-linguitic, meta-level feature.

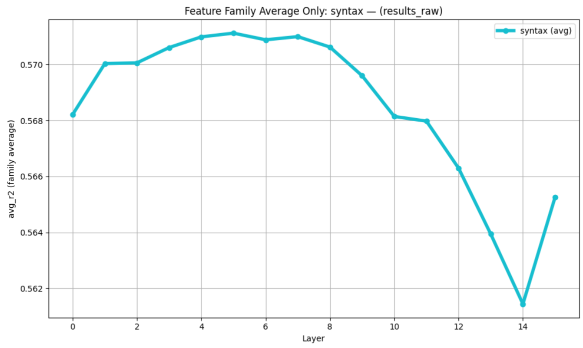


Figure 13: Llama-1B raw MLP: trend of scores for `syntax` feature

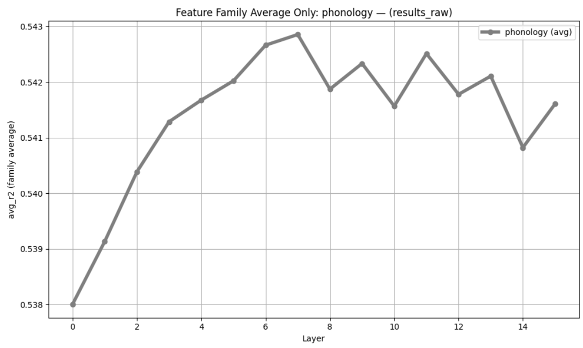


Figure 14: Llama-1B raw MLP: trend of scores for `phonology` feature

4.2 Layerwise pattern: R^2 peaks mid-layer or decays with depth

For both the SAE and raw MLP neurons, middle layers encode maximal language identity (strongest typology signal). Insome cases the discriminative power concentrates in the early layers (e.g fam for both Llama-1B and Gemma-1B) and sometimes in the middle layer (e.g syntax for Llama-1B). We also see a peculiar case for phonology being concentrated in the final layers for LLama-1B.

Final layers shift toward semantic representation and next-token prediction, suppressing typological distinctions.

4.3 SAEs yield consistently higher R^2 than raw activations

SAE representations are more disentangled, more axis-aligned, and less superposed. Typological signals are isolated more cleanly, improving linear recoverability.

Language-identity features appear to be inherently sparse in the latent space.

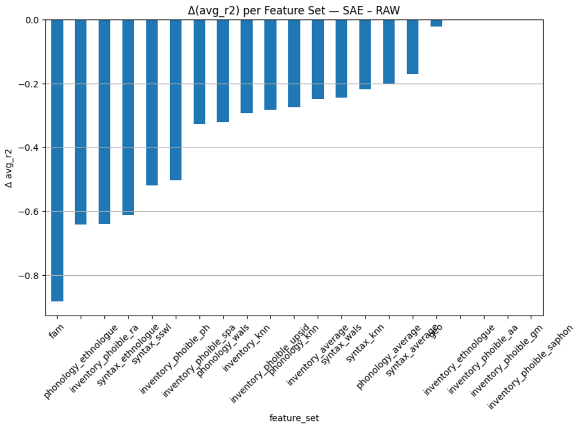


Figure 15: Differences between average scores for the SAE activations and raw MLP activations for Llama-1B

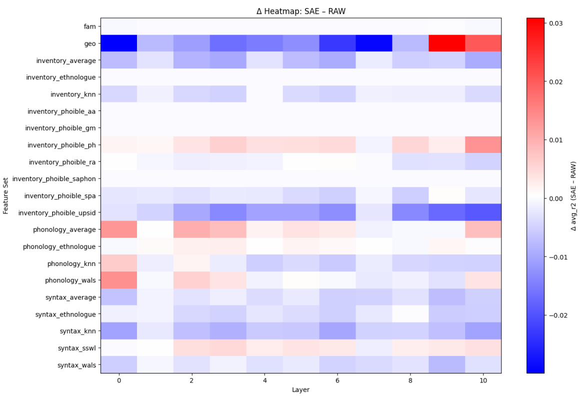


Figure 16: Layerwise differences between average scores for the SAE activations and raw MLP activations for Gemma-2B

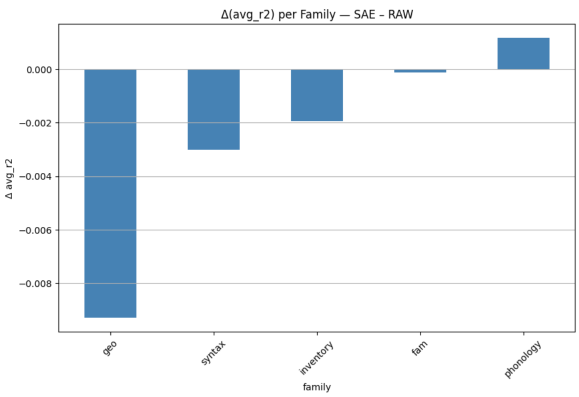


Figure 17: Differences between average scores for the SAE activations and raw MLP activations for Gemma-2B, on a feature-family level

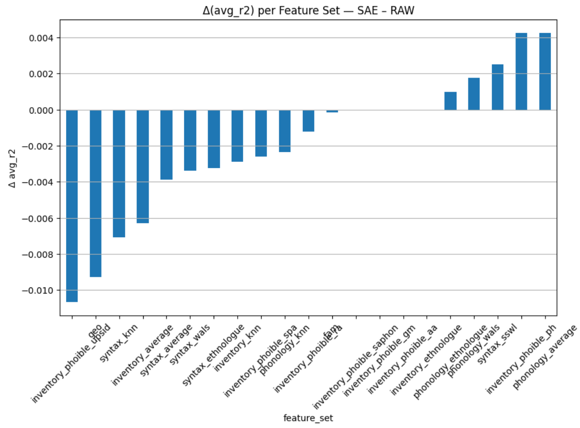


Figure 18: Differences between average scores for the SAE activations and raw MLP activations for Gemma-2B, on a feature-set level

4.4 Gemma 2B > LLaMA 1B in typology recoverability

Gemma-2B's Larger model capacity (2B vs 1B), better multilingual tokenizer, and more diverse training data lead to stronger and more linearly accessible typology information.

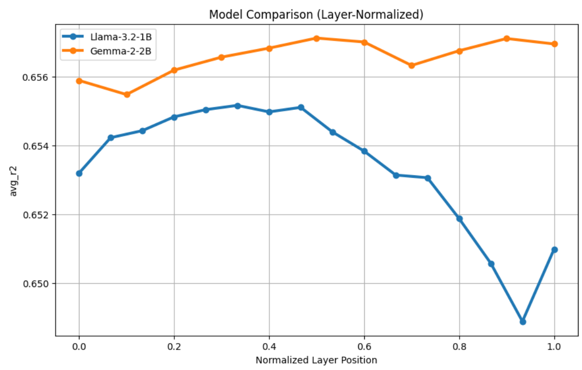


Figure 19: Gemma-2B has higher average scores along a normalized layer axis as compared to Llama-1B

4.5 Shared neuron typological correlations

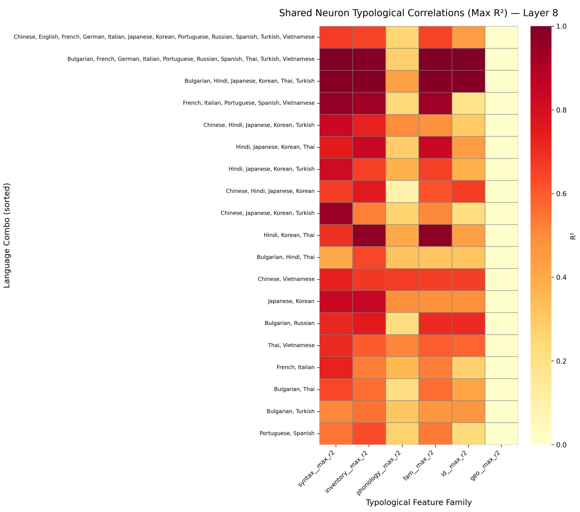


Figure 20: Shared neuron typological correlations

4.6 Activation Steering

5 Discussion

Our updated analysis uncovers a critical limitation of language-specific neuron approaches: they largely capture orthographic artifacts, not script-invariant linguistic abstractions. On the positive side, inventory and family features correlate strongly with the identified neurons, suggesting that these features are not arbitrary. In particular, we see that related languages (same family) tend to share many neurons and that turning on a family's neurons can improve model performance on that family's languages. This implies that the neurons do reflect some consistent groupings among languages.

The drastic drop from syntax $\rightarrow$ phonology $\rightarrow$ geography shows that:

- Syntax is moderately learned but not robust.
- Phonology is poorly represented (as expected).
- Geographic variation (which often correlates with shared linguistic diffusion areas) is almost totally missing.

As a consequence, models excel at script, orthography, morphology, but fail at long-range structural rules, morphosyntactic hierarchy, cultural/topological context

This may explain why multilingual models struggle with low-resource languages, syntactically divergent languages, morphologically rich languages.

However, the romanization findings reveal that these neurons are fragile with respect to script changes. In linguistic terms, the neurons are sensitive to the "orthographic envelope" of the language: their activations are tied to the character-level patterns of a script.

The near-zero overlap (Jaccard similarity tending to 0) means that **even when a language's phonology is preserved (via transliteration), the model's internal representation treats it as effectively a different language**.

This orthography-dependence has practical consequences. It suggests caution in using these neurons for cross-lingual steering or transfer. For example, one might hope to activate "Hindi neurons" to boost Hindi output, but if the input were in romanized Hindi (as is common on the Internet), those neurons would not fire. Similarly, multilingual text augmentation or transliteration-based transfer could break the alignment. In effect, the model is not learning an abstract concept like "this is Hindi semantics"; it is learning "this sequence of Devanagari characters signals Hindi." Our results therefore raise concerns about claims of language-specific features being truly language-universal, and highlight the importance of testing such claims.

Multilingual LMs rely too heavily on superficial, family-level similarities and insufficiently encode the structural properties necessary to align distant or low-resource languages. Typology signals are present but superposed, weak, and vanish in deeper layers, causing unstable and inequitable cross-lingual transfer.

5.1 What Do Multilingual Activations Reveal About Typology and Transfer?

Our probing results uncover a structured picture of how multilingual language models encode typological information. We first confirm several observations reported in prior work, but more importantly, we identify multiple previously unreported phenomena with implications for cross-lingual transfer.

Established patterns we confirm. Consistent with earlier findings on mBERT and XLM-R, we observe that (i) language identity is represented most strongly in middle transformer layers; (ii) larger models tend to encode typological information more robustly; and (iii) coarse genealogical features (e.g., language family) are easier to recover than fine-grained syntactic or phonological traits. Our results also support prior conclusions that non-linguistic factors such as geographic proximity leave almost no recoverable signal.

A hierarchical structure of typological recoverability. Across models, layers, and activation types, we find a remarkably stable hierarchy: *Family* > *Inventory* > *Syntax* > *Phonology* > *Geography*. This cross-model consistency has not been documented previously. It suggests that multilingual representations are dominated by coarse genealogical and orthographic regularities, while deep structural properties are encoded only weakly. This hierarchy provides a unified view of what multilingual models "pay attention to" when differentiating languages.

SAEs expose hidden typological structure. One of our most striking findings is that sparse autoencoder (SAE) activations yield substantially higher R^2 scores than raw MLP activations across *almost all* typological categories. This reveals that typological signals are not absent from raw representations; rather, they are heavily superposed and become more linearly accessible after SAE-based disentanglement. To our knowledge, this is the first demonstration that SAEs can surface typological structure in multilingual models, suggesting that superposition is a key bottleneck in cross-lingual representation quality.

Architectural differences matter: Gemma 2B vs. LLaMA 1B. While scaling trends are known, direct comparisons across model *families* at similar parameter counts are scarce. We find that Gemma 2B exhibits uniformly higher typological recoverability than LLaMA 1B—especially in early layers—which we attribute to improved tokenizer coverage, more balanced multilingual data, and architectural refinements such as RMSNorm scaling. These results highlight that multilingual alignment quality is not solely a function of model size, but also of tokenizer design and pretraining mixture.

Layerwise dynamics reveal semantic washout. We observe two distinct layerwise patterns: (i) an inverted-U curve where typological signals peak mid-model and decay in deeper layers, and (ii) monotonic decay where early layers are typology-dominated. The latter is especially prominent in the SAE latents of Gemma 2B, suggesting that certain architectures encode language identity extremely early. The "late-layer decay" supports the hypothesis that deeper layers prioritize task-specific semantic abstraction, causing typological information to be suppressed.

Implications for cross-lingual transfer. Taken together, these findings offer mechanistic explanations for why cross-lingual transfer degrades for distant or low-resource languages. Models over-rely on genealogical and surface-level cues, encode syntax and morphology only weakly, and experience loss of typological distinctions in deeper layers. Furthermore, the heavy superposition revealed by SAEs indicates that typological structure exists but is difficult for the model to use reliably, especially in low-resource or typologically atypical languages. These structural limitations point to an emerging picture of "shallow multilingualism," where models align languages primarily through surface statistics rather than deeper structural properties.

References

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